High-Rate Full-Hand Tactile Sensing for Sim-to-Real Recognition and Grasping

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***Abstract*— Full-hand tactile sensing provides contact information that vision can hardly capture after a dexterous hand encloses an object. This is important for stable grasping and shape recognition under occlusion. This paper presents a full-hand tactile perception framework for sim-to-real dexterous grasping. The platform uses a four-finger dexterous hand with 16 degrees of freedom. It is equipped with an in-house, low-cost piezoresistive tactile sensing system that is densely integrated across the entire hand. It also provides stable high-rate tactile signals at 300 Hz. In simulation, MuJoCo is used to match the positions of different tactile sensing taxels. Ray-normal projection and local spacing correction are used to align the tactile layout on the hand surface. The simulated tactile responses are then mapped to the real voltage domain through force-voltage calibration. A multi-patch tactile classifier is trained and evaluated with simulated and real data. In experiments, the model achieves 73.33% accuracy on a real dataset with 150 object-grasp samples. After fine-tuning with another 150 real samples, the accuracy reaches 92.67% on the test set. These results show that high-rate full-hand piezoresistive sensing, geometric taxel alignment, and calibrated tactile simulation can support practical grasp-based object shape recognition.**

I. INTRODUCTION

Human manipulation relies strongly on both vision and touch. In dexterous grasping, vision guides approach and closure, but once the hand envelops an object, visual observations alone often cannot support fine manipulation or reliable object-property perception. Tactile sensing directly reflects contact force, pressure distribution, grasp stability, and local geometry. It therefore remains central to contact-rich manipulation [1]–[3].

However, this capability places strict requirements on the sensing interface of a dexterous hand. Many optical and vision-based tactile sensors provide rich local measurements. Their optical stacks, housings, or mechanical stiffness still make dense coverage over fingertips, finger links, and palms difficult [4]. Custom hand geometries also make commercial tactile modules expensive to adapt. At the simulation level, existing tactile simulators and benchmarks still focus mainly on fingertip contacts, vision-based tactile sensors, or local modules. As a result, geometric alignment of full-hand tactile units and sim-to-real transfer for whole-hand tactile skins remain insufficiently addressed, while collecting large-scale real tactile grasp data remains costly.

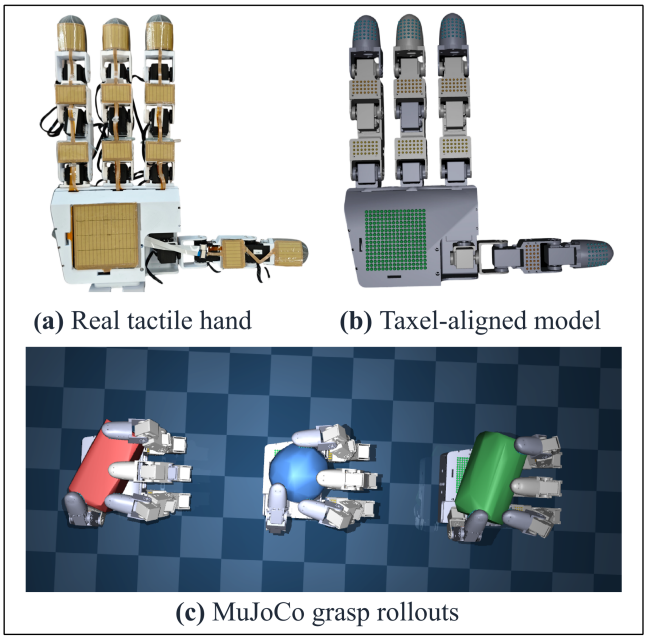


Fig. 1. Overview of the tactile grasp recognition platform. **(a)** Real FPC tactile hand. **(b)** Taxel-aligned MuJoCo hand. **(c)** Simulated fixed-grasp rollouts for box, sphere, and cylinder objects.

To address these challenges, we present a full-hand tactile perception framework for fixed-grasp object recognition (Fig. 1). Here, fixed-grasp means that the hand closing trajectory is fixed, while object size, pose, orientation, and local surface perturbations vary. On the hardware side, dense flexible printed-circuit (FPC) piezoresistive patches provide modular, replaceable, and easily integrated full-hand coverage. On the simulation side, a MuJoCo tactile model maps taxels onto the three-dimensional hand surface by ray-normal projection and local spacing correction. The simulator converts contact depth to voltage-like tactile maps, so a classifier can be trained on simulated grasps and tested on real tactile grasping, thus achieving zero-shot learning.

In summary, the main contributions are:

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This work was supported by the Natural Science Foundation of Top Talent of SZTU (Grant No. GDRC202411) and the SZTU-NU Information Technology Partner Exchange Program.

- We develop a highly integrated and extensible tactile sensing system for a robotic hand. The hand platform integrates 620 FPC-based piezoresistive taxels across the fingertips, finger links, and palm. With multi-channel Serial Peripheral Interface (SPI) acquisition, Direct Memory Access (DMA) double buffering, and a unified frame format, the system delivers a stable 300 Hz full-hand tactile stream. A parameterized sensor interface supports modules with different taxel counts, dimensions, and data-transfer lengths.
- We propose a taxel-aligned sim-to-real grasp perception framework. It registers simulated taxels to the MuJoCo hand surface to maintain geometric consistency with the real tactile skin, converts simulated contact depth to voltage-like tactile maps, trains a multi-patch classifier on simulated fixed-grasp episodes, and validates zero-shot and adapted recognition on real tactile data.

II. RELATED WORK

Full-hand piezoresistive tactile perception for dexterous grasp recognition requires both high-rate tactile hardware and calibrated sim-to-real learning. We briefly review related work from these two aspects and motivate the proposed full-hand tactile acquisition and fixed-path grasp recognition framework.

A. Piezoresistive Tactile Sensing for Dexterous Hands

Dexterous hands provide the mechanical basis for contact-rich manipulation. Our previous studies developed a dexterous-hand platform and investigated tactile-based grasp stability assessment [5], [6]. In contrast, this work focuses on high-rate full-hand tactile acquisition and tactile object-shape recognition after grasp closure.

Tactile sensing has been widely applied to object recognition, contact estimation, grasp control, and manipulation feedback [1]–[3], [7]–[10]. Vision-based tactile sensors provide rich local deformation information, but their cameras, optical components, and packaging hinder dense integration over fingertips, finger links, and palms [4]. Recent studies have developed 3-D-printed piezoresistive arrays, capacitive tactile arrays, replaceable tactile skins, and dense visuo-tactile systems [11]–[14]. However, these systems mainly address sensor fabrication, local contact sensing, material recognition, replaceability, or visuo-tactile policy learning. High-rate acquisition and unified region-wise organization of full-hand piezoresistive signals for grasp recognition have received less attention.

Full-hand piezoresistive sensing is also affected by nonlinear force-voltage responses, saturation, hysteresis, taxel variation, and baseline drift. These effects motivate calibrated tactile representations for learning.

B. Taxel-Aligned Simulation and Zero-Shot Learning

Simulation reduces the cost of collecting real tactile grasp data, while domain randomization is widely used to narrow the sim-to-real gap [15]. TACTO, Tactile Gym 2.0, and SimTacLS have demonstrated tactile simulation and sim-to-real learning for vision-based tactile sensors [16]–[18]. Recent studies have also explored zero-shot sim-to-real policy transfer with magnet-based tactile sensors and self-supervised perception for tactile-skin-covered dexterous hands [19], [20]. However, most existing work focuses on vision-based or fingertip tactile sensors, magnet-based sensing, or policy transfer, and does not directly address sim-to-real transfer of multi-patch full-hand piezoresistive voltage fields.

For full-hand piezoresistive sensing, the sim-to-real gap mainly arises from mismatches in geometric layout and signal response. The simulator must preserve the shape, spatial location, and arrangement of each taxel while mapping penetration depth or contact force to nonlinear voltage signals consistent with real sensor outputs. Existing studies rarely address both issues within a unified full-hand tactile framework. We therefore formulate full-hand tactile sim-to-real recognition as a joint problem of taxel alignment and signal-response calibration, as detailed in the next section.

III. METHOD

The proposed system has two coupled parts. The first part is the real tactile hand, which supplies high-rate full-hand voltage fields. The second part is the sim-to-real perception pipeline, which generates aligned simulated tactile data and trains a tactile-only classifier. Both parts use the same 12-patch representation, so a grasp sample has the same structure in simulation and on hardware.

TABLE I

FULL-HAND TACTILE SYSTEM SPECIFICATIONS.

Item	Specification
Hand actuation	16 active DoF, fully actuated
Hand structure	4 fingers, 16 joints, 3-D printed links
Tactile modality	In-house FPC piezoresistive array
Sensing layout	12 patches, 620 physical taxels
Patch distribution	1 palm, 4 fingertip, and 7 finger-link patches
Palm array	16×16 taxels
Fingertip array	6×7 taxels per fingertip
Finger-link array	4×7 taxels per link patch
Acquisition buses	3-channel SPI
Buffering	DMA double buffer
Frame size	632 bytes
Stable frame rate	300 Hz at 2 Mbps
Recognition window	5 stable tactile frames

A. High-Rate Full-Hand Piezoresistive Tactile Hand

The real platform is a four-finger dexterous hand with 16 active degrees of freedom. The palm, fingertips, and finger links are covered with in-house FPC-based piezoresistive tactile arrays. Each tactile array consists of a top electrode layer, a pressure-sensitive layer, and a bottom electrode layer. Its thin structure allows it to be attached to 3D-printed finger links and the palm surface. Compared with vision-based tactile sensors, the piezoresistive array does not require cameras, lighting units, or thick optical packaging. It is therefore more suitable for large-area coverage on compact dexterous hands.

The full-hand tactile skin is divided into 12 logical tactile patches, covering the main contact regions of the palm,

thumb, index finger, middle finger, and ring finger. Each patch contains multiple taxels, and each taxel corresponds to an independent piezoresistive sensing unit. Table I summarizes the main parameters of the tactile system, including the patch layout, taxel number, array size, and acquisition settings. The patches mainly differ in physical location, boundary shape, and contact function, rather than sensing resolution. Therefore, this work preserves a patch-wise data organization. This keeps the learning input consistent with the physical hand structure and provides a basis for simulated taxel registration and multi-region tactile feature fusion.

High-rate acquisition is important for fixed-path grasp recognition. During hand closure, the contact state between the fingers and the object changes rapidly. If the tactile readout is too slow, transient pressure distributions during contact formation may be missed. It may also reduce the amount of useful tactile information available shortly after grasp closure. Existing tactile arrays and tactile skins have shown the value of thin sensors for robotic hands [11]–[14]. However, these works mainly focus on sensor fabrication, local contact sensing, material recognition, replaceable tactile skins, or visuo-tactile policy learning. In contrast, this work focuses on high-rate acquisition, unified frame organization, and scalable integration for full-hand piezoresistive tactile arrays.

Fig. 2 shows the full-hand tactile acquisition architecture. The system adopts a two-level acquisition scheme. The first level is performed on the tactile sensing boards. Each sensing board corresponds to one functional hand region, such as one finger or the palm, and independently scans the tactile array in that region. The sensing board continuously updates local tactile frames through double buffering to maintain data freshness. The second level is performed on the main controller. The STM32F407 main controller reads multiple sensing boards in parallel through three SPI buses and uses DMA for SPI reception and Universal Asynchronous Receiver/Transmitter (UART) transmission. The three SPI buses are assigned to sensing boards according to the actual system configuration, avoiding the need to read all tactile data through a single serial path. The main controller also uses a double-buffered frame queue, allowing one frame to be captured while another frame is transmitted to the host through UART. This structure overlaps tactile scanning, controller readout, and data transmission, thereby reducing the full-hand frame acquisition time.

The acquisition software is parameterized to support tactile module expansion. Each tactile module is defined by its SPI bus, chip-select pin, frame length, frame offset, enable flag, and sensor ID. During initialization, the program automatically computes the frame offset of each enabled sensor according to its frame length and assigns the enabled sensors to SPI1, SPI2, and SPI3 according to their SPI handles. Therefore, when a tactile patch is added or replaced, only parameters such as the sensor length, enable flag, chip-select pin, and SPI assignment need to be modified. The acquisition scheduler and unified frame parser remain unchanged.

Finally, the system outputs full-hand tactile data in a unified frame format. Each frame contains a frame header and data segments from different tactile modules. Under a 2 Mbps communication setting, the system provides stable full-hand tactile frames at 300 Hz. It was determined from the configured transmission rate and frame-output settings, while continuous software monitoring of frame sequence numbers and transmitted/received frame counts showed no incomplete or dropped frames. This high-rate tactile data stream can quickly provide stable contact information after the dexterous hand grasps an object, offering reliable input for subsequent object shape recognition.

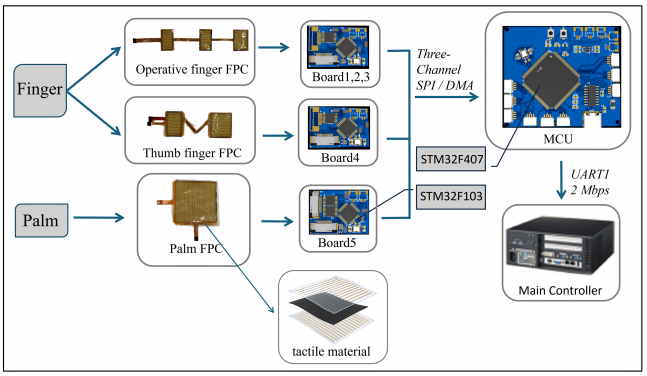


Fig. 2. Full-hand tactile acquisition architecture. Multiple tactile sensor arrays are sampled by distributed acquisition boards and transmitted to the main controller through a three-channel SPI/DMA interface and a high-rate UART link.

B. Taxel-Aligned Tactile Simulation and Zero-Shot Sim-to-Real Learning

The goal of this subsection is to make simulated and real fixed-grasp samples usable by the same tactile classifier. Because only the closing trajectory is fixed and the classifier receives post-contact tactile frames, the main problem is data compatibility rather than action selection. We therefore align the two domains in geometry and signal. Geometry alignment keeps each virtual taxel in the same patch and array position as its real counterpart. Signal alignment converts MuJoCo contact geometry to the voltage-like domain of the piezoresistive skin. We call this conversion depth–force–voltage (D-F-V) mapping: simulated penetration depth is converted to normal force, then to voltage, and finally to a full-hand normalized frame. Fig. 3 shows the complete path to the shared classifier input.

1) Shared tactile representation. We first fix the classifier input. One grasp sample contains the five stable tactile frames saved after the hand stops:

$$X = \{X_p\}_{p=1}^{P}, \qquad X_p \in \mathbb{R}^{T\times H_p\times W_p}, \tag{1}$$

where $P = 12$, $T = 5$, and X_p is the sequence of patch p. The patch sizes H_p, W_p may differ across palm, finger-link, and fingertip regions, but patch names and ordering are fixed. The label is $y \in \{\text{box}, \text{sphere}, \text{cylinder}\}$. Thus simulated and real episodes can be read by the same loader and classifier.

2) Virtual taxel placement. The simulator needs a 3-D position for every virtual taxel. Starting from the real patch topology, we place a regular 2-D grid on each patch and project each grid seed onto the hand mesh by ray-mesh

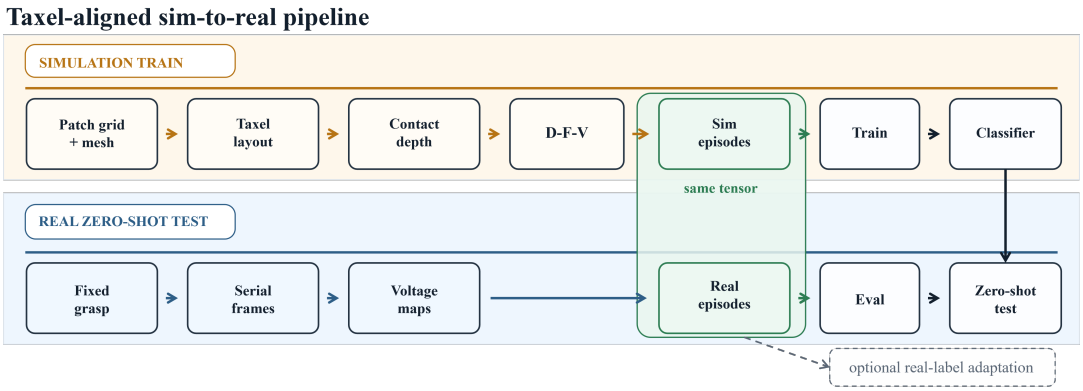


Fig. 3. Framework of taxel-aligned tactile simulation and zero-shot sim-to-real learning. The simulation stream projects tactile patch grids to the hand mesh, queries contact depth, and converts it to normalized voltage-like maps through depth–force–voltage calibration. The real stream executes the same fixed grasp path and stores normalized voltage maps. Both streams use the same 12-patch, five-frame tensor interface.

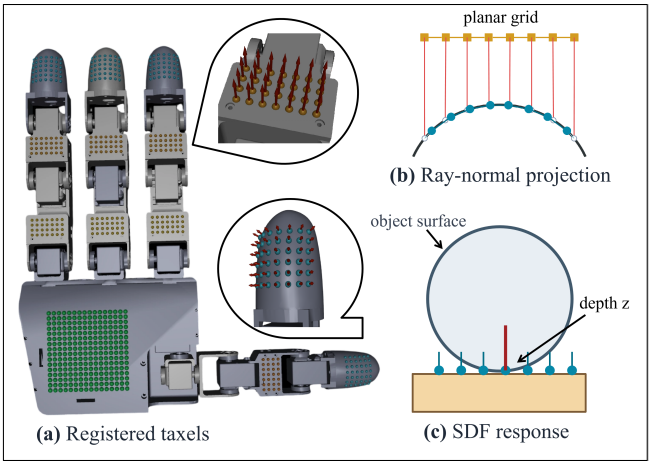


Fig. 4. Detailed taxel registration and tactile response generation. Virtual taxels are projected to the hand mesh, locally relaxed on curved surfaces, and queried against object geometry to obtain penetration depth.

intersection. For taxel (i, j) in patch p, the projection is

$$\begin{aligned} r_{pij} &= c_p + u_j e_p^u + v_i e_p^v + \delta n_p, \\ q_{pij} &= \Pi_{\mathcal{M}}\left(r_{pij}, -n_p\right), \end{aligned} \tag{2}$$

The resulting registration and contact-query process is illustrated in Fig. 4.

Here all quantities are expressed in SI units unless noted. $c_p \in \mathbb{R}^3$ is the seed center of patch p (m); $e_p^u, e_p^v \in \mathbb{R}^3$ are unit directions of the local u and v grid axes (dimensionless); $u_j \in \mathbb{R}$ and $v_i \in \mathbb{R}$ are the local metric coordinates (m) of grid node (i, j) along e_p^u and e_p^v, respectively, so that $u_j e_p^u + v_i e_p^v$ places the taxel in the patch tangent plane; $n_p \in \mathbb{R}^3$ is the outward unit normal of the patch (dimensionless); and $\delta > 0$ is a small offset (m) that moves the ray origin slightly outside the surface to stabilize intersection. The operator $\Pi_{\mathcal{M}}(\cdot, -n_p)$ returns the first intersection of the ray with the hand mesh $\mathcal{M}$, and $q_{pij} \in \mathbb{R}^3$ is the registered taxel position (m). Eq. (2) therefore maps a flat taxel grid onto the real hand surface. Curved fingertip points are then locally relaxed to preserve neighboring spacing. The output is a MuJoCo hand model with tactile sites and a layout file storing patch names, taxel indices, rows, columns, and display orientation.

3) Simulated tactile signal. With the registered layout, the simulator runs fixed-grasp rollouts for box, sphere, and cylinder objects. Object size, pose, yaw, and local surface perturbation are randomized, while the closing path is fixed. At each frame, taxels outside the object produce zero response, and taxels inside the object produce a positive penetration depth. Fingertip taxels also check nearby MuJoCo contact points to reduce missed small contacts.

The D-F-V mapping converts this geometric contact measure into the normalized voltage-like signal used by the classifier. The mapping is obtained from two calibration procedures. For the real force–voltage calibration, a small rigid probe applies controlled normal loads to representative physical taxels. The corresponding voltage outputs are recorded to construct a shared force–voltage table $h(\cdot)$ over approximately 0.05–$1.0\,\mathrm{N}$, as illustrated in Fig. 5(c). For the simulated depth–force calibration, the same probe procedure is reproduced on a representative palm taxel in MuJoCo. A total of 24 normal-force levels from 0.1 to $10.0\,\mathrm{N}$ are applied, and the corresponding settled penetration depths are recorded to construct a shared depth–force table $g(\cdot)$, as illustrated in Fig. 5(f). Both calibration tables are shared across all 620 taxels. This choice is supported by the multi-point consistency results and is consistent with the subsequent full-hand normalization. Residual taxel-response variations are mitigated through training-time tactile domain randomization and fine-tuning on real-robot data. The simulated force–depth curve defines a monotone map $g(\cdot) : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ that converts penetration depth (m) to normal force (N) by piecewise-linear interpolation of the calibrated depth–force table. The real force–voltage curve defines $h(\cdot) : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ that converts force (N) to voltage (V) by piecewise-linear interpolation of the calibrated force–voltage table. During grasp simulation, each taxel response is converted through penetration depth, force, and voltage before normalization:

$$\begin{aligned} z_i &= \max\left(-d_\Omega(q_i), 0\right), \\ F_i &= g(z_i), \qquad \tilde{V}_i = h(F_i), \\ V_i &= \frac{\tilde{V}_i - \min_j \tilde{V}_j}{\max_j \tilde{V}_j - \min_j \tilde{V}_j + \varepsilon}. \end{aligned} \tag{3}$$

In Eq. (3), Ω denotes the grasped object geometry, and $d_\Omega(q_i)$ is the signed distance (m) from the taxel position $q_i \in \mathbb{R}^3$ to the object surface, where $d_\Omega(q_i) < 0$ indicates penetration into Ω. Thus, $z_i = \max(-d_\Omega(q_i), 0)$ is the non-negative

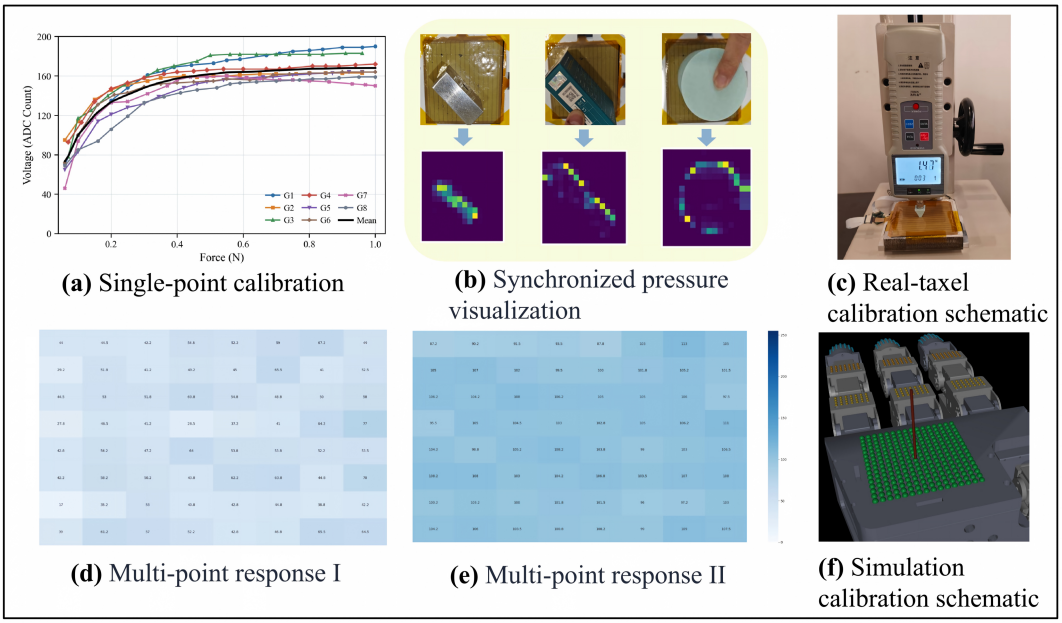


Fig. 5. Sensor characterization and depth–force–voltage calibration. **(a)** Force–voltage calibration at eight representative taxel positions. **(b)** Synchronized physical contacts and tactile maps. **(c)** Real force–voltage calibration setup for the piezoresistive sensor. **(d)–(e)** Multi-point loading responses. **(f)** Simulated depth–force calibration for converting penetration depth to contact force.

penetration depth (m), $F_i = g(z_i)$ is the estimated normal contact force (N), and $\tilde{V}_i = h(F_i)$ is the corresponding voltage-like response (V). The index j runs over all active taxels of the full hand in the current frame, so the last line performs full-hand min–max normalization and produces the dimensionless response $V_i \in [0, 1]$. The constant $\varepsilon > 0$ prevents division by zero when the frame responses are nearly uniform. When the stopping condition is reached, five stable frames are saved as one simulated episode.

4) Real formatting and learning. The real hand uses the same fixed closing path and stop-then-save rule. A serial frame is parsed into the 12 patch order used by the simulator. ADC values are converted to voltages; optional baseline subtraction, clipping, and full-hand min–max normalization are then applied as in Eq. (3). The program stores the same five-frame patch tensors and a manifest entry, so real and simulated samples share patch names, frame count, tensor convention, and label space.

The classifier input is the 12-patch tensor dictionary X in Eq. (1), and the output is a three-class shape prediction over box, sphere, and cylinder. We use a supervised multi-patch CNN. For each patch, the five stable frames are stacked as $T = 5$ input channels of a lightweight 2-D CNN encoder that is *not* weight-shared across patches: two 3×3 convolutions (channels $5 \to C_h \to C_o$) with ReLU activations, a 2×2 max-pool after the first convolution, and global average pooling, yielding a C_o-dimensional patch embedding. Unless otherwise stated, non-palm patches use $C_h = 24$ and $C_o = 48$, while the larger palm encoder uses $C_h = 32$ and the same $C_o = 48$. Optional Squeeze-and-Excitation is applied inside each encoder. The 12 patch embeddings are stacked to a sequence in $\mathbb{R}^{12\times C_o}$, modulated by a learnable per-patch gate, refined by $L = 2$ layers of multi-head self-attention (4 heads) for inter-patch context, then flattened and passed to an MLP head $\mathbb{R}^{12C_o} \to 96 \to 48 \to 3$ with LayerNorm, GELU, and dropout (0.3). Training minimizes cross-entropy with AdamW (learning rate 3×10^{-4}, weight decay 10^{-4}), batch size 16, and up to 30 epochs with cosine-style learning-rate scheduling and early stopping on a held-out simulated validation split. During simulation pretraining we use whole-sample global_std normalization and strong tactile domain randomization (signal scale/gain jitter, nonlinear intensity, noise, palm dimming/dropout, and occasional full-patch dropout) so that the model relies on spatial contact patterns rather than absolute voltage magnitude. The simulator provides 3000 labeled episodes (1000 per class). In zero-shot testing, the simulation-trained weights are frozen and evaluated on real episodes. For real-data adaptation, the same checkpoint is fine-tuned on the 150 labeled real training trials with a lower learning rate 5×10^{-5}, batch size 8, up to 60 epochs, light augmentation, balanced class weights, and unfrozen encoders, then evaluated on the disjoint 150-trial real test set.

The method therefore leads to two experimental checks. The hardware must first provide fast and stable full-hand tactile frames. Then the taxel-aligned simulation and learning pipeline must recognize object shape from real fixed-grasp samples. The experiments follow this order.

IV. EXPERIMENTS AND RESULTS

The experiments follow the two parts of the method. Sec. IV-A checks whether the tactile hardware produces stable high-rate full-hand signals. Sec. IV-B checks whether the taxel-aligned simulation and zero-shot learning pipeline transfers to real fixed-grasp shape recognition.

A. Characterization of the Piezoresistive Tactile Sensor

To evaluate the basic sensing performance of the proposed piezoresistive tactile array, we tested its single-point force

response, contact-shape perception, and multi-point response consistency. Fig. 5 shows the sensor responses under different test conditions.

Fig. 5(a) shows the ADC readings of single taxels under different normal forces. Multiple taxels were selected for repeated calibration, and the mean response was calculated. The results show that the sensor output increases monotonically with force. In the low-force range of 0.1–0.5 N, the response shows an approximately linear trend. This indicates that the tactile unit can effectively capture pressure changes during light contact. As the force further increases, the response gradually becomes slower, which is consistent with the nonlinear and saturation characteristics of piezoresistive materials.

Fig. 5(b) shows synchronized pressure visualization under different contact shapes. The metal block, ruler edge, and circular object produce different pressure distributions. The high-response regions in the heatmaps are consistent with the actual contact contours. This result shows that the tactile array can capture not only contact intensity but also local contact shape and spatial distribution.

Fig. 5(d) and Fig. 5(e) further evaluate the response consistency under multi-point loading. Under the same pressure level, the average responses of different regions are close to each other, indicating stable sensing performance across the array. Fig. 5(d) and Fig. 5(e) correspond to two different pressure levels. As the external pressure increases, the overall response level increases accordingly, while the spatial distribution remains relatively uniform.

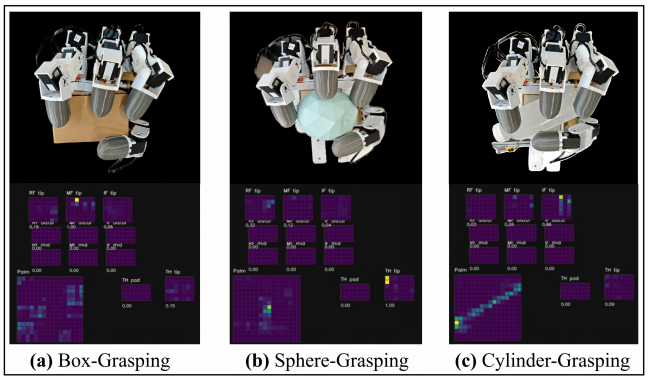
These results show that the proposed piezoresistive tactile array provides stable low-force response, clear contact-shape representation, and consistent multi-point feedback. These properties provide a reliable sensing basis for subsequent full-hand tactile acquisition and grasp-based object shape recognition.

B. Fixed-Grasp Shape Recognition

The task uses three primitive shape classes: box, sphere, and cylinder. The simulated training set has 3000 D-F-V episodes, 1000 per class. The real dataset contains 300 fixed-grasp trials from five object instances per class and 20 grasps per instance. The 300 trials are split into two non-overlapping 150-trial subsets for real training and testing, with 50 trials per class in each subset; no real trial appears in both. Each trial provides five stable full-hand tactile frames after the hand stops. Fig. 6 shows representative real grasp examples and tactile maps after closure.

Table II reports accuracy and macro-F1 on the same 150-trial real test set. All methods use the training-time tactile augmentation described in Sec. III-B. D-F-V denotes the depth-force-voltage mapping in Eq. (3). The D-F-V zero-shot model is trained only on simulated data and reaches 73.33% accuracy. After real-data adaptation, the same architecture reaches 92.67% accuracy and 92.67% macro-F1.

The ablations clarify the component roles. Fingertip-only D-F-V gives 40.00% accuracy because many contacts occur on the palm and finger links. This suggests that fixed-grasp shape classification relies more on spatial activation patterns than precise signal magnitudes. Thus, D-F-V mainly supplies physical consistency for sim-to-real data, whereas full-hand coverage and real-data adaptation drive most of the accuracy gain.



(a) Box-Grasping **(b)** Sphere-Grasping **(c)** Cylinder-Grasping

Fig. 6. Fixed-grasp examples and tactile maps after closure.

TABLE II

FIXED-GRASP RECOGNITION ON THE BALANCED 150-TRIAL REAL TEST SET. D-F-V DENOTES DEPTH-FORCE-VOLTAGE.

Method	N	Acc.	Macro-F1
D-F-V, zero-shot	150	73.33%	73.47%
D-F-V + real adaptation	150	**92.67%**	**92.67%**
Fingertip-only D-F-V	150	40.00%	40.00%

Class-level results after real adaptation are balanced: box, sphere, and cylinder reach 90%, 92%, and 96% accuracy, respectively. Most remaining errors occur between box and cylinder trials, whose edge or side contacts can look similar under the fixed path.

Discussion and limitations. The system classifies simple object shapes from a short tactile window after a fixed grasp, but it does not estimate 6-D pose, material, slip, or grasp stability. It has also only been tested on three primitive classes. Future work should extend the object set, evaluate cross-session sensor drift, and add closed-loop tactile adjustment.

V. CONCLUSIONS

This paper presents a full-hand piezoresistive tactile system and a taxel-aligned sim-to-real framework for fixed-grasp object-shape recognition. The hand integrates 620 tactile taxels and streams complete tactile frames at 300 Hz. The simulator aligns virtual taxels to the hand mesh, converts penetration depth through D-F-V calibration, and generates scalable training data for a multi-patch tactile classifier. On a balanced real test set, calibration-aware zero-shot transfer reaches 73.33% accuracy, and real-data adaptation reaches 92.67%. These results show that full-hand tactile coverage, high-rate acquisition, and calibrated tactile simulation are useful building blocks for deployable dexterous-hand perception.

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